

i When to Use

Use Pearson's correlation when you have two quantitative (continuous) variables and want to test whether there is a linear relationship between them.

H Research Hypothesis: Over time, the age gap between the actors playing Lois Lane and Clark Kent has decreased, such that older portrayals will have larger age differences.

H Null Hypothesis: There is no linear relationship between year and the age difference between the actors playing Lois Lane and Clark Kent.

Running a Correlation

1. **Analyze → Correlate → Bivariate**
2. Highlight each of the desired variables and click the arrow button
3. Be sure "Pearson" and "Two-tailed" buttons are checked
4. Click "Options" — check "Means and standard deviations"
5. Click "OK" to run the analysis

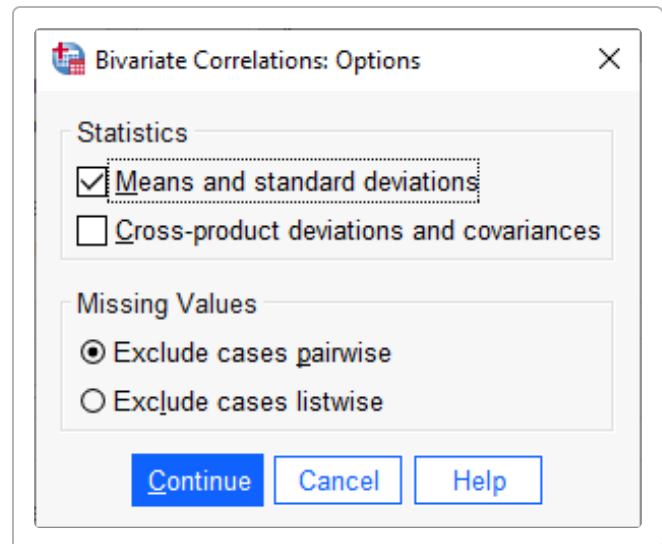
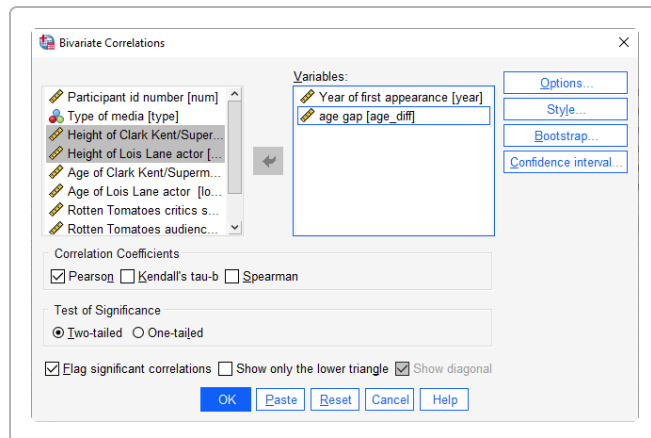
SPSS Syntax

```
CORRELATIONS
/VARIABLES=year age_diff
/PRINT=TWOTAIL
/STATISTICS DESCRIPTIVES
/MISSING=LISTWISE.
```

- ① List the variables to correlate
- ② Get 2-tailed p-values (use **ONETAIL** for directional hypothesis)
- ③ Get means and standard deviations
- ④ Use listwise deletion (use **PAIRWISE** to retain more cases)

i Note

The **standard deviation (SD)** tells about the expected variation of individual scores around the mean of the distribution.



SPSS Output

Descriptive Statistics

	N	Mean	Std. Deviation	Std. Error	Minimum	Maximum
year	11	1992.09	25.47	7.68	1948.00	2025.00
age_diff	9	5.72	4.14	1.38	1.15	13.03
Valid N (listwise)	9					

Correlations

		year	age_diff
year	Pearson Correlation	1.000	-.584
	Sig. (2-tailed)		.099
	N	9	9
age_diff	Pearson Correlation	-.584	1.000
	Sig. (2-tailed)	.099	
	N	9	9

**. Correlation is significant at the 0.01 level (2-tailed).

*. Correlation is significant at the 0.05 level (2-tailed).

The correlation of -0.584 indicates that there is a significant negative linear relationship between these variables — as year increases, the age difference tends to decrease.

SPSS reports the sample size ($N = 9$), but you should report the degrees of freedom ($df = N - 2 = 7$).

Reporting the Results

Variable	Descriptive Statistics			Correlations
	M	SD	n	1
1. Year of portrayal	1992.09	25.47	11	—
2. Age difference (Lois - Clark)	5.72	4.14	9	-0.58

Note. * $p < .05$. ** $p < .01$. *** $p < .001$.

Important: SPSS reports the sample size (N), but you should report the degrees of freedom ($df = N - 2$). Even if the printout shows $p = .000$, never report $p = .000$ — instead report “ $p < .001$ ”.

SPSS presents a “correlation matrix” — notice:

- The correlation between a variable and itself is perfect (1.00)
- The matrix is symmetrical (upper right and lower left correlations are identical)

It is important to report the univariate statistics for the variables before presenting the correlation and the significance test. Often these are presented in a table.

As in the example, be sure to communicate:

- The research hypothesis (if there is one)
- The statistical results
- Whether or not those results support the research hypothesis

Example Write-Up

Sample APA Write-Up

There was no significant linear relationship between Year of Portrayal ($M = 1992.09$, $SD = 25.47$) and Age Difference ($M = 5.72$, $SD = 4.14$), $r(7) = -.58$, $p = .099$. This result does not support the research hypothesis that the age gap decreases over time.

When reporting correlations, describe the results so that a reader with limited statistical expertise will understand how the variables are related. Saying “Year and age difference are significantly negatively correlated” is correct, but less informative than the phrasing in the examples.